



Job title: Research Assistant - Grand Challenges Living Lab

Job reference number: B02-09864

Professional Expertise: Research and Research Support

Department: Experimental Psychology (F67)

Location: London

Working Pattern: Full time

UCL Salary: £36,433-£41,833 per annum (pro-rata for part-time)

Job Type: Fixed-term

Is a DBS required for the role: No

Working Type: Hybrid

Working Hours per Week: 36.5

Available for Secondment: No

Closing date: 24/12/2025

About us

The Department of Experimental Psychology, within the Division of Psychology and Language Sciences (PaLS), offers a world leading research and education environment aimed at understanding the psychological and biological basis of behaviour. PaLS is consistently ranked in the top 10 centres for psychology in the world and UCL is the top-ranked university in the UK for research power in Psychology, Psychiatry and Neuroscience according to the UK's Research Excellence Framework 2021. The Department of Computer Science, within the Faculty of Engineering, is home to some of the world's most influential and creative researchers. It is harnessing that power to bring real-world benefits to everyone, and equipping the next generation of computer scientists to tackle society's biggest challenges. An exciting opportunity has arisen to join a multidisciplinary team to develop predictive models of energy usage in order to cut costs and emissions, enhancing the sustainability of UCL.

About the role

We wish to appoint a Research Assistant to contribute to a UCL-sponsored research project which aims to develop models to forecast energy demand at UCL estates to cut costs and emissions. The project is a collaboration between UCL Experimental Psychology, UCL Computer Science, and UCL The Bartlett Centre for Advanced Spatial Analysis. The research assistant will be jointly supervised by Professor Maarten Speekenbrink (UCL Experimental Psychology), Professor Benjamin Guedj (UCL Computer Science) and Dr Carlo Ciliberto (UCL Computer Science). The successful applicant will develop a predictive model of energy usage patterns across locations on UCL campus. Using advanced probabilistic machine learning techniques, they will develop a robust forecasting model of short- and medium-term energy usage across UCL estates. The model will consider seasonal and other periodic fluctuations, space characteristics (e.g. building size, occupancy, specialist equipment, etc.) to build a normative model of past and future energy usage for different locations. The model will allow real-time monitoring and prediction, as well as intelligent anomaly detection (e.g. higher use than expected from similar locations on UCL campus) to inform decision-making and planning. Using principles of compositional kernel methods (e.g. Gaussian Processes), the model will decompose usage patterns into components which allow for more straightforward explanation of results and generalisation of predictions across locations.

The position is for 9 months on a full-time basis. We are looking for a person with relevant experience in probabilistic modelling of time-series data and applications to large data sets. The supervisory team will provide additional support and guidance on model development and application.

We will consider applications to work on a part-time, flexible and job share basis wherever possible. This role is eligible for hybrid working with a minimum of 60% on site. This appointment is subject to UCL Terms and Conditions of Service for Research and Professional Services Staff. Please visit <https://www.ucl.ac.uk/human-resources/conditions-service-research-teaching-and-professional-services-staff> for more information.

About you

The successful candidate must have a University degree (BSc/MSc/MRes) in relevant area (e.g. machine learning, statistics, artificial intelligence). They must have a deep understanding of Gaussian process regression or other probabilistic models for time-series data and experience with major Python packages for machine learning and probabilistic modelling (e.g. scikit-learn, tensorflow, torch, pyMC). They will have excellent Python programming skills and an ability to work both independently and collaboratively. They will be able to manage their time and work to deadlines and demonstrate excellent interpersonal, oral and written communication skills. They will have the ability to work as part of a multidisciplinary team, and collaborate with other researchers. They will have a commitment to academic research and the highest ethical and professional standards in research and education. They will be highly motivated and hard-working, and ideally have an interest in sustainability and energy reduction.

What we offer

As well as the exciting opportunities this role presents, we also offer some great benefits some of which are below:

- 41 Days holiday (27 days annual leave 8 bank holiday and 6 closure days)
- Additional 5 days' annual leave purchase scheme
- Defined benefit career average revalued earnings pension scheme (CARE)
- Cycle to work scheme and season ticket loan
- Immigration loan
- Relocation scheme for certain posts
- On-Site nursery
- On-site gym
- Enhanced maternity, paternity and adoption pay
- Employee assistance programme: Staff Support Service
- Discounted medical insurance

Visit <https://www.ucl.ac.uk/work-at-ucl/rewards-and-benefits> to find out more. For job board postings use the condensed plain text: As well as the exciting opportunities this role presents, we also offer great benefits. Please visit <https://www.ucl.ac.uk/work-at-ucl/rewards-and-benefits> to find out more.

Our commitment to Equality, Diversity and Inclusion

As London's Global University, we know diversity fosters creativity and innovation, and we want our community to represent the diversity of the world's talent. We are committed to equality of opportunity, to being fair and inclusive, and to being a place where we all belong. We therefore particularly encourage applications from candidates who are likely to be underrepresented in UCL's workforce. These include people from Black, Asian and ethnic minority backgrounds; disabled people; and for our Grade 9 and 10 roles, women. Athena Swan Status Statement • Our department holds an Athena SWAN Silver award, in recognition of our commitment and demonstrable impact in advancing gender equality. You can read more about our commitment to Equality, Diversity and Inclusion here: <https://www.ucl.ac.uk/equality-diversity-inclusion/>

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